Software Engineering lab 01
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Problem statement number : 61
Household Carbon Footprint & Target Tracker

REQUIREMENTS TABLE

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh), gas consumption, and transport (km) logs to compute total monthly carbon equivalent (CO_2e) emissions.	**High**	**Pass:** For valid monthly electricity, gas, and transport inputs, the calculated CO_2e matches the specified conversion-factor formulas. **Fail:** The system produces an incorrect or negative CO_2e value.	Core functionality for measuring household carbon footprint.
FR-002	Functional	The system shall allow a resident to enter and update monthly household utility and transport consumption data.	**High**	**Pass:** A resident can save valid electricity, gas, and transport values for a selected month and retrieve the saved values. **Fail:** Valid values are not saved or retrieved correctly.	Enables residents to maintain accurate monthly consumption records.
FR-003	Functional	The system shall display the household's monthly CO_2e emissions and historical emission trends.	**High**	**Pass:** After selecting a month or date range, the system displays the corresponding CO_2e values and a historical comparison graph. **Fail:** The graph does not display the stored emission values correctly.	Helps residents understand changes in their carbon footprint over time.

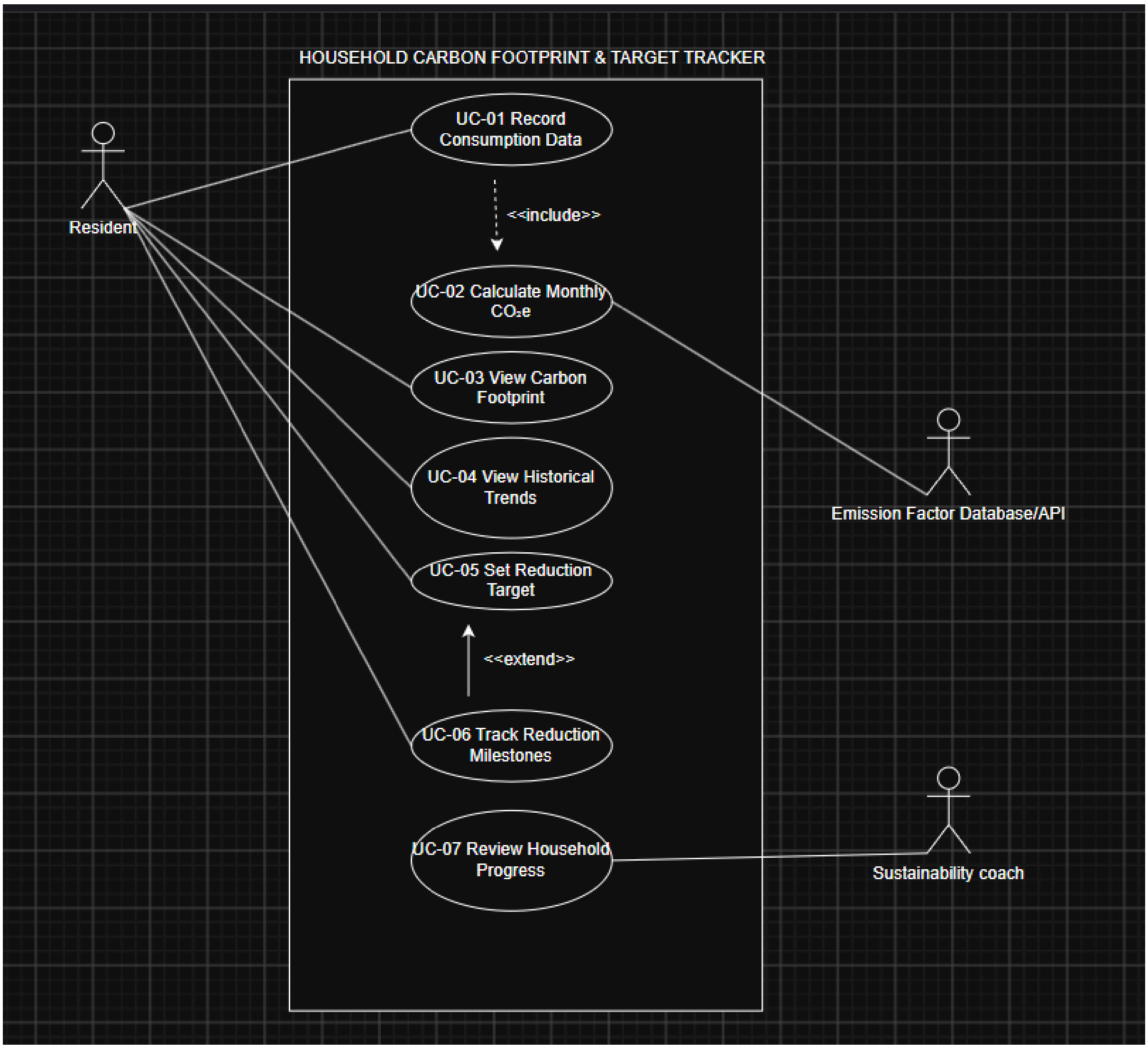
FR-004	Functional	The system shall allow a resident to set an annual carbon-reduction target and track progress toward the target.	**High**	**Pass:** When a resident sets a valid annual reduction target, the system displays the target and current progress percentage. **Fail:** The target or progress is not displayed correctly.	Supports measurable carbon-reduction planning.
FR-005	Functional	The system shall notify the resident when a defined carbon-reduction milestone is achieved or missed.	**Medium**	**Pass:** When the household reaches or fails to reach a configured milestone, the system generates the corresponding notification. **Fail:** No notification is generated when the milestone condition is satisfied.	Encourages residents to monitor and improve their sustainability performance.
NFR-001	Nonfunctional – Performance & Security	The annual carbon reduction milestone chart shall render interactive historical comparison graphs in under 200 ms.	**High**	**Pass:** Benchmarking tests confirm graph rendering completes in **< 200 ms** under simulated peak load while meeting the specified security standards. **Fail:** Rendering exceeds 200 ms or security requirements are violated.	Ensures fast and secure visualization of historical carbon data.
NFR-002	Nonfunctional – Usability	The system shall allow a resident to record a month's consumption data using no more than five input actions after opening the data-entry form.	**Medium**	**Pass:** A usability test confirms that a resident can enter and submit valid monthly consumption data within **5 input actions**. **Fail:** More than five input actions are required.	Makes monthly data entry simple and convenient for residents.

USE CASES

ID	Use Case

UC-01	Record Consumption Data
UC-02	Calculate Monthly CO_2e
UC-03	View Carbon Footprint
UC-04	View Historical Trends
UC-05	Set Reduction Target
UC-06	Track Reduction Milestones
UC-07	Review Household Progress

Use case diagram



USE CASE FLOW

Use Case: UC-01 — Record Consumption Data

Primary Actor

Resident

Supporting Actor

Emission Factor Database/API

Preconditions

1. The resident has access to the Household Carbon Footprint & Target Tracker.
2. The resident is authenticated in the system.
3. The system has access to the required emission conversion factors.

Postconditions

1. The monthly electricity, gas, and transport data is stored successfully.
2. The system calculates the household's monthly CO_2e emissions.
3. The updated carbon footprint is available for historical tracking.

Main Success Scenario

1. Resident selects **"Record Consumption Data"**.

2. System displays the monthly data-entry form.

3. Resident selects the month and enters electricity consumption in kWh, gas consumption, and transport distance in km.

4. System validates that all entered values are valid and non-negative.

5. System retrieves the required emission conversion factors from the Emission Factor Database/API.

6. System calculates the total monthly CO_2e emissions using the entered consumption values and conversion factors.

7. System stores the monthly consumption data and calculated CO_2e value.

8. System displays a confirmation message containing the recorded monthly CO_2e emissions.

9. Use case ends successfully.

Alternate Flow

A1 — Invalid Consumption Data

4a. If the resident enters an invalid or negative consumption value, the system displays an error message identifying the invalid field.

4b. Resident corrects the invalid value.

4c. System validates the corrected values again.

4d. If all values are valid, the system continues from Step 5 of the main success scenario.

A2 — Emission Factor Database Unavailable

5a. If the Emission Factor Database/API is unavailable, the system displays a message indicating that CO_2e calculation cannot currently be completed.

5b. The system does not finalize the carbon calculation.

5c. Resident is prompted to try again later.

5d. Use case ends without successfully calculating CO_2e.